

Insert 14 to the binary tree



Date: ...../...../.....

fit@hcmus

Start at the root:

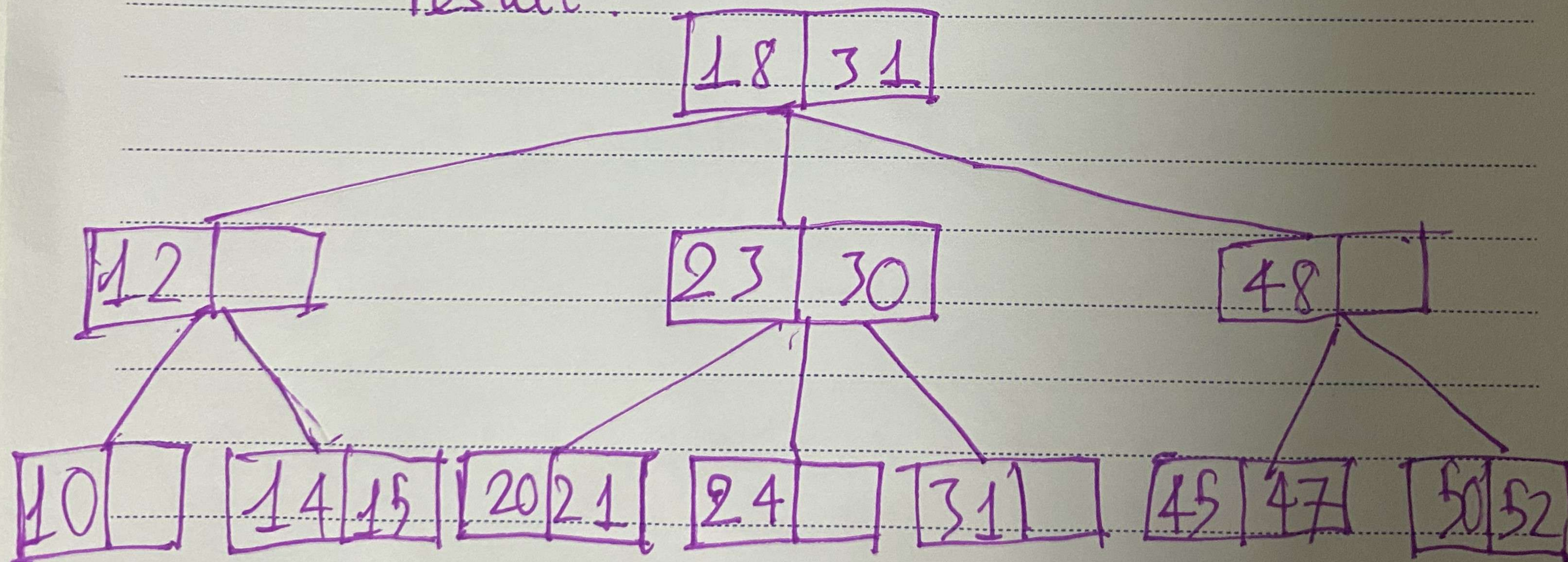
|    |    |
|----|----|
| 18 | 31 |
|----|----|

$14 < 18 \Rightarrow 14$  go down to the left

|    |  |
|----|--|
| 12 |  |
|----|--|

$14 > 18$  and the right child node can afford more value  
 $\Rightarrow 14$  go down to the right

result:





Insert 55 to the binary tree



Date: ..... / ..... / .....

Start at the root:

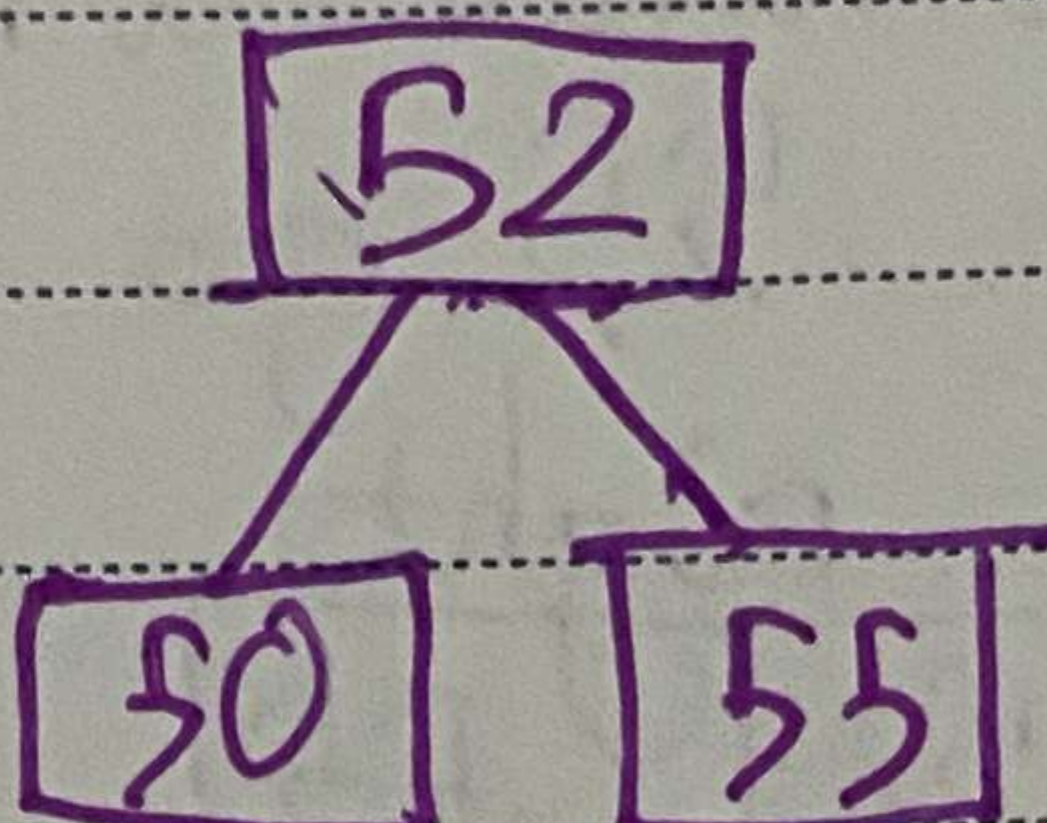
18 | 31 : 55 &gt; 31

 $\Rightarrow$  55 go down to the right

48 | : 55 &gt; 48

 $\Rightarrow$  55 go down to the right

50 | 52 | 55

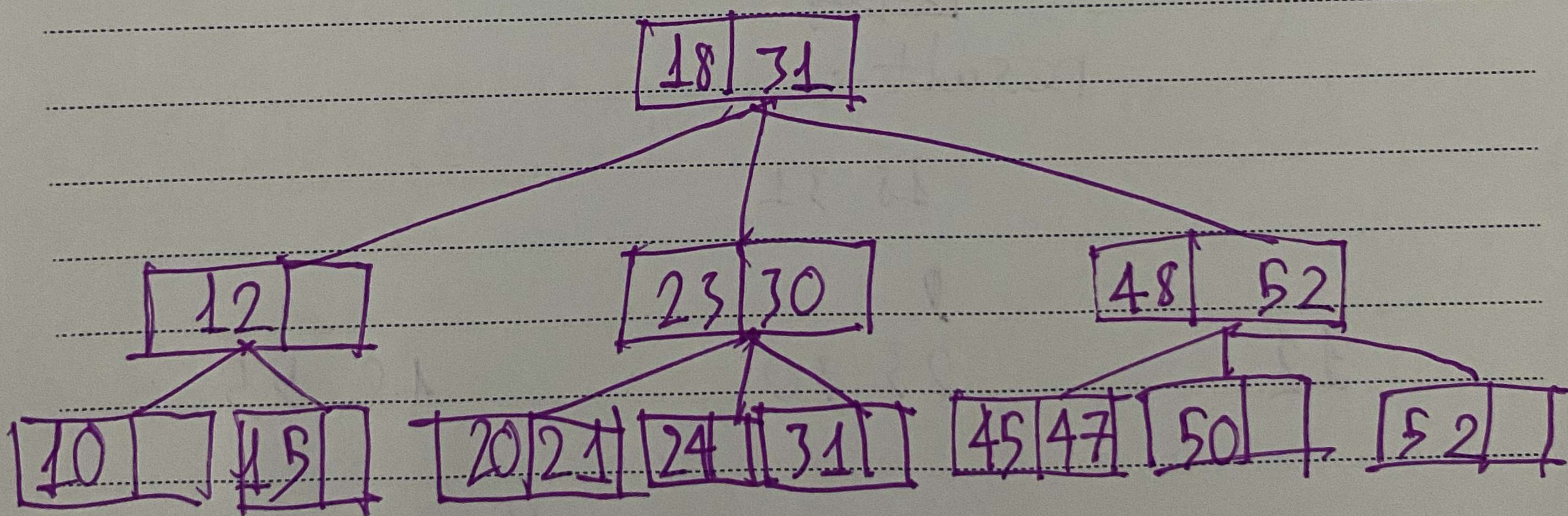
 $\rightarrow$ 

[48]

can afford 1 more value

 $\rightarrow$ 

final result





Insert 19 to the binary tree



Start at the root:

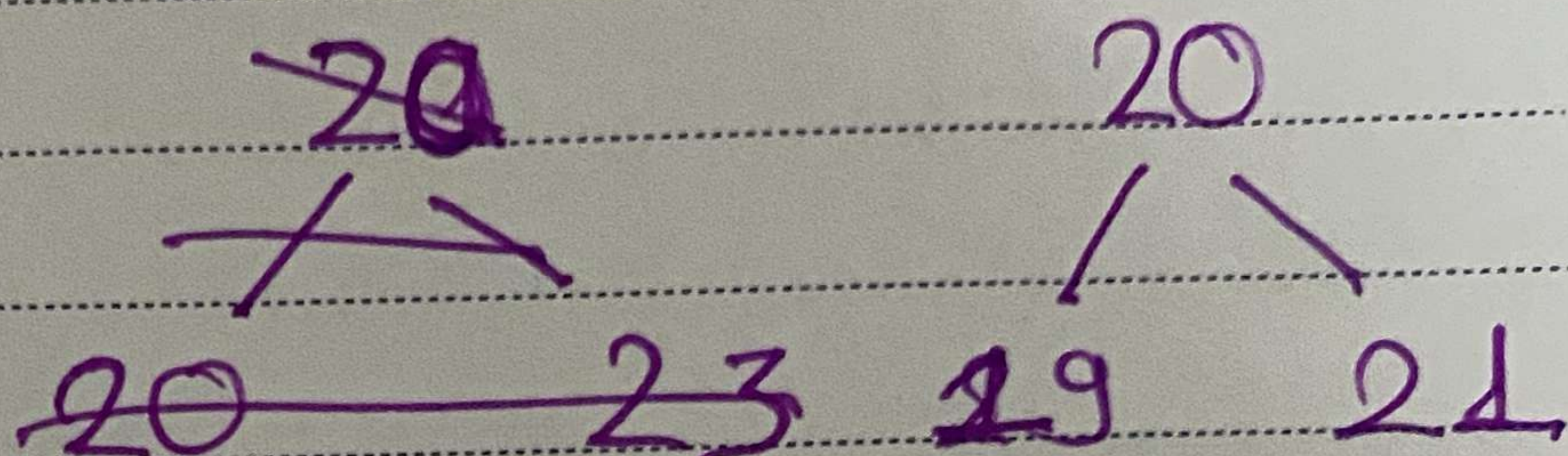
$$18 | 31: \quad 18 < 19 < 31$$

$\Rightarrow 19$  go down to the middle

$$23 | 30: \quad 19 < 23$$

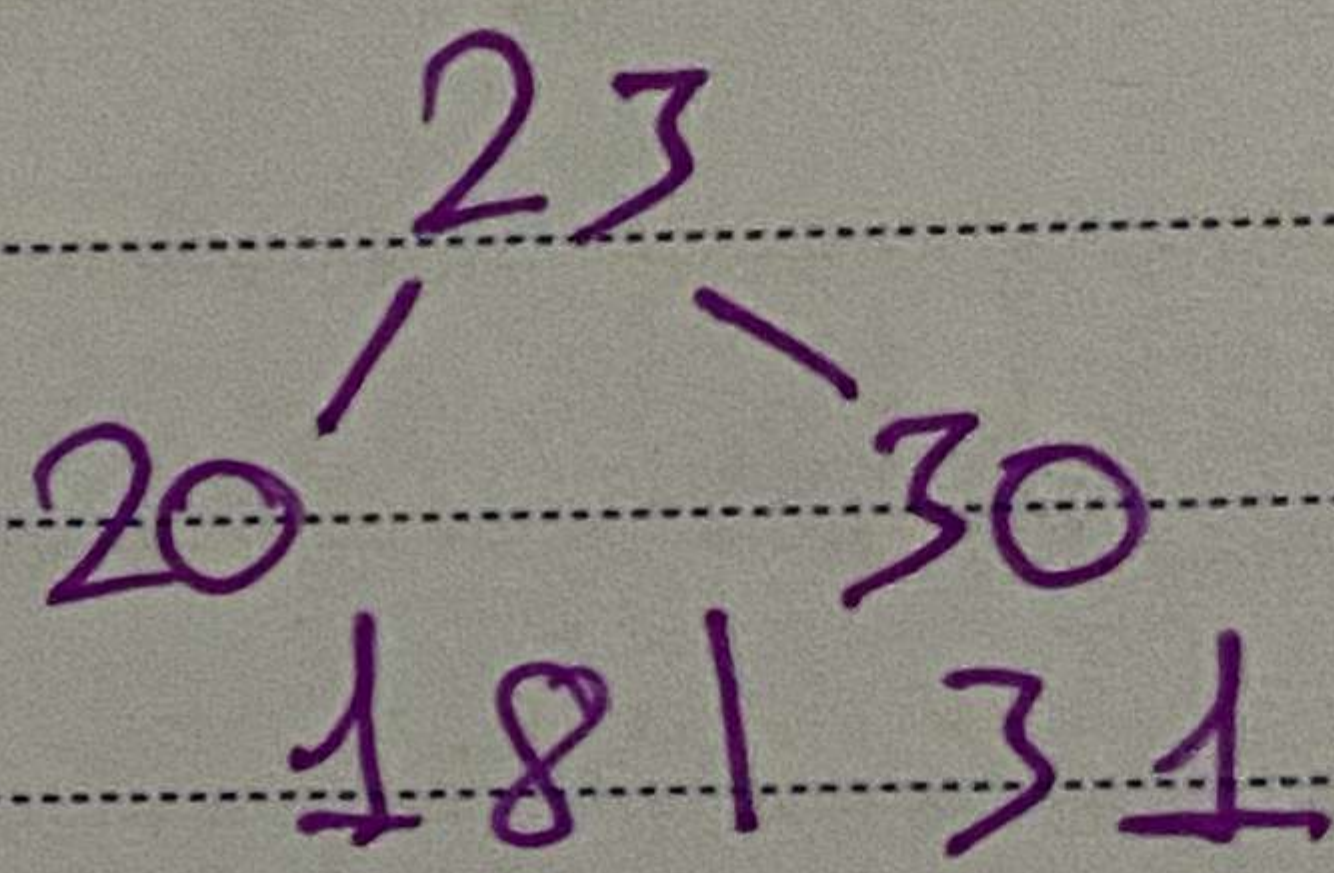
$\Rightarrow 19$  go down to the left

$$20 | 21 | 19 \rightarrow$$



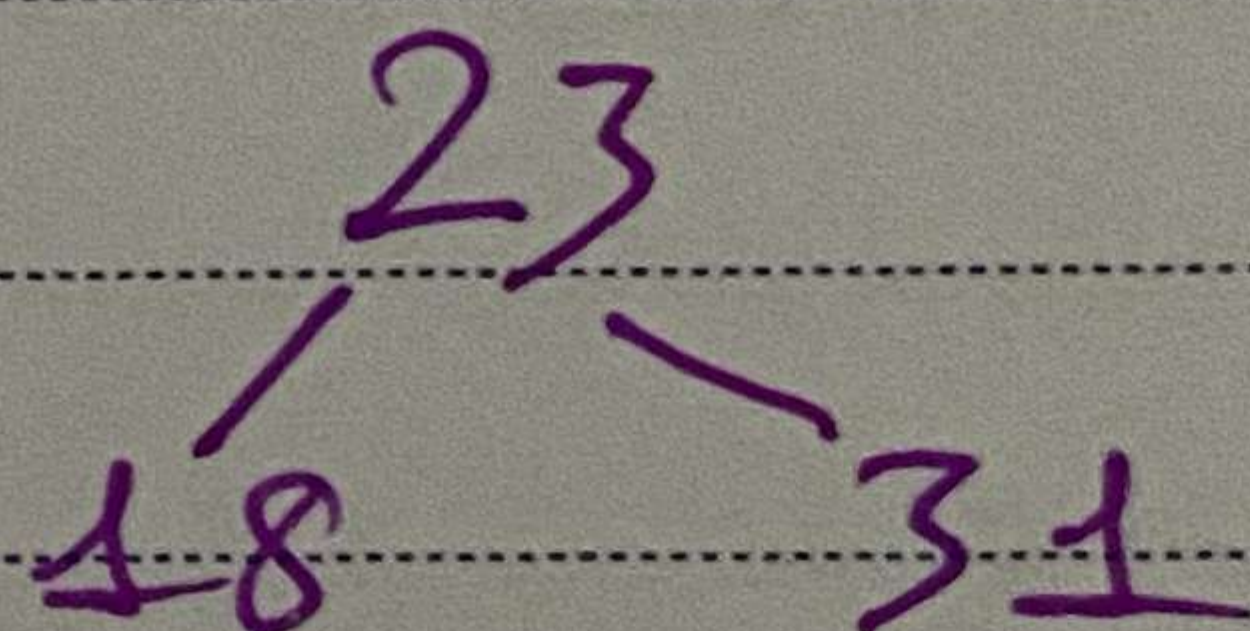
$\rightarrow 20$  go up to  $23 | 30$

$$20 | 23 | 30 \rightarrow$$



$\rightarrow 23$  go up to

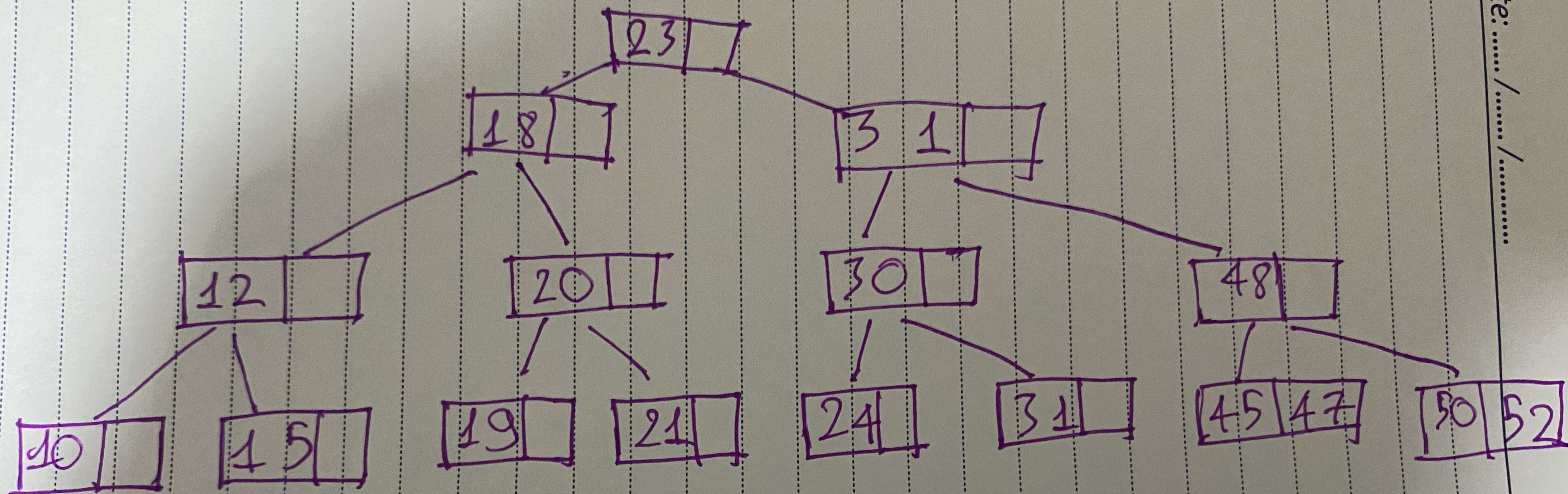
$$18 | 23 | 31 \rightarrow$$





Date: ..... / ..... / .....

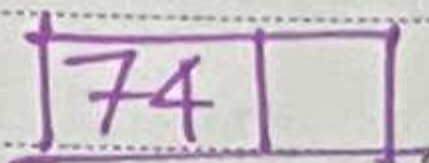
final result



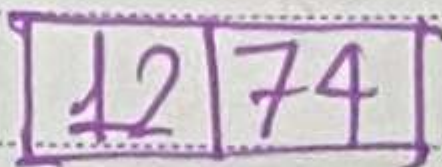


Insert to 2-3 tree

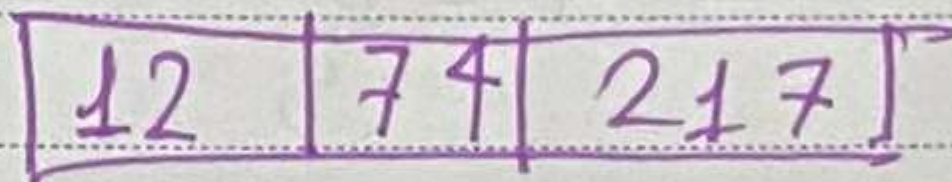
74 →



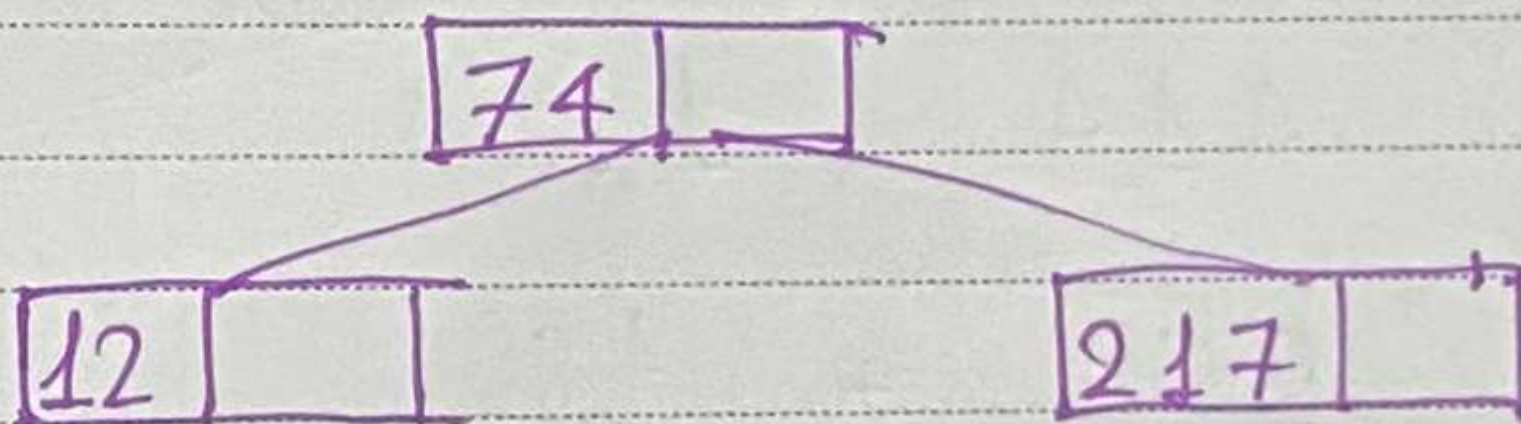
12 →



217 →

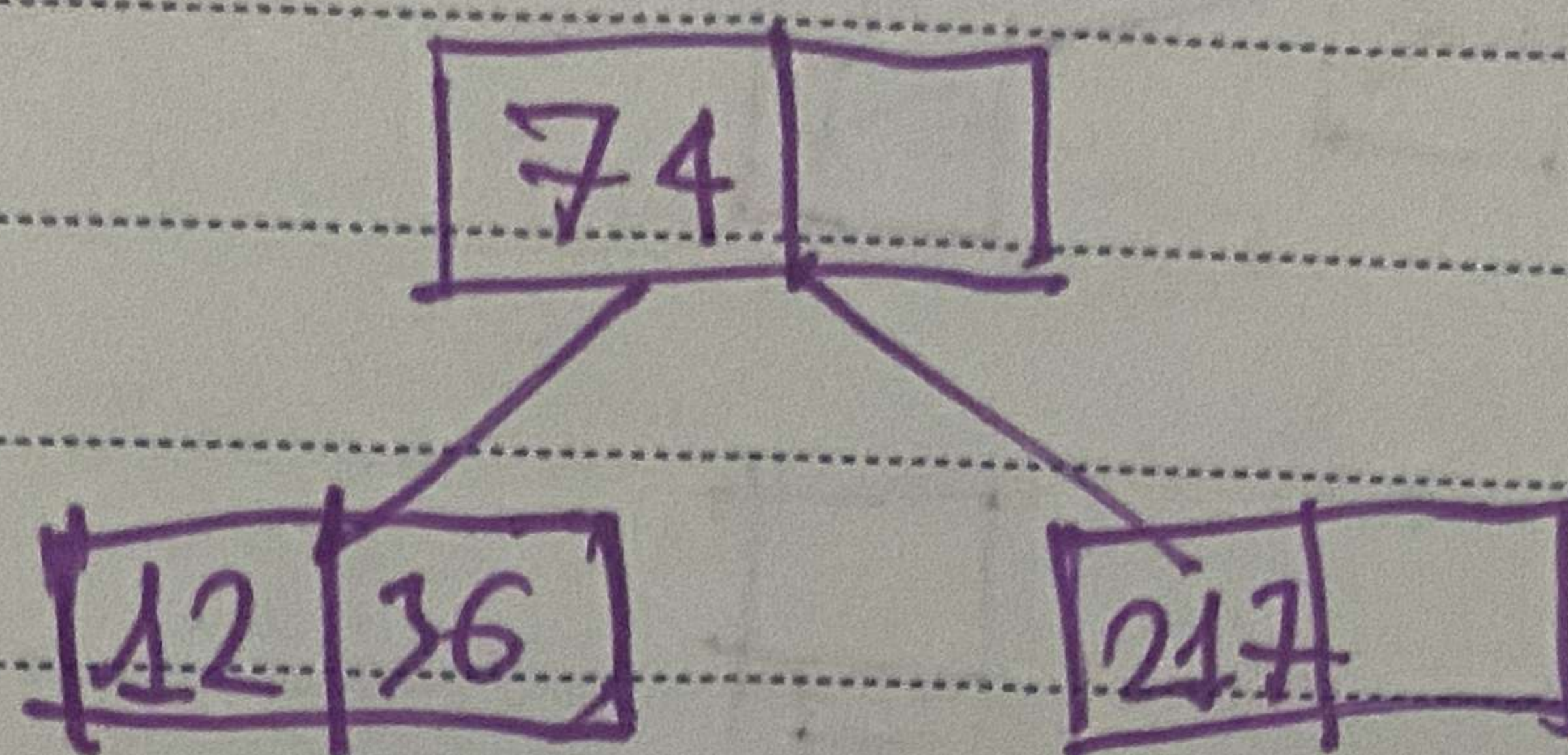


→

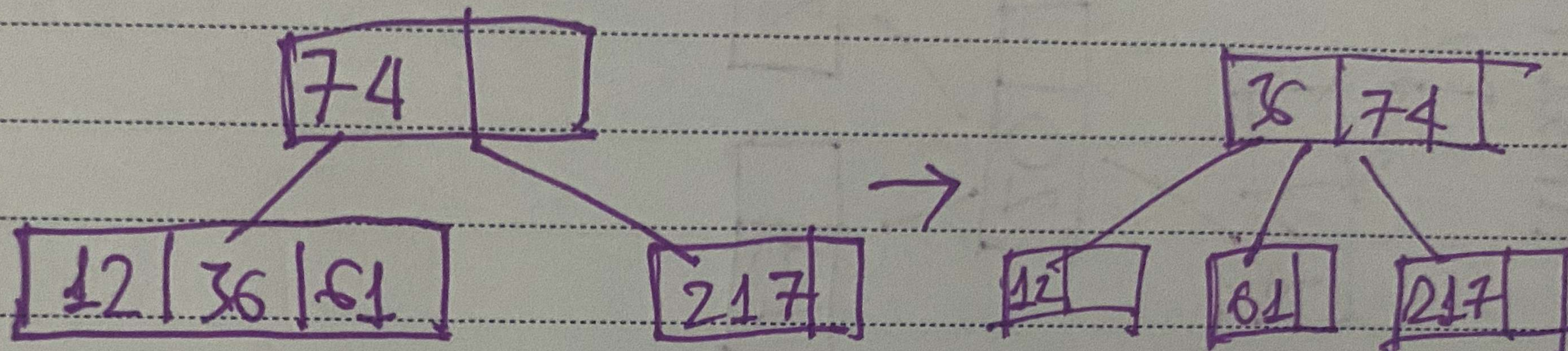




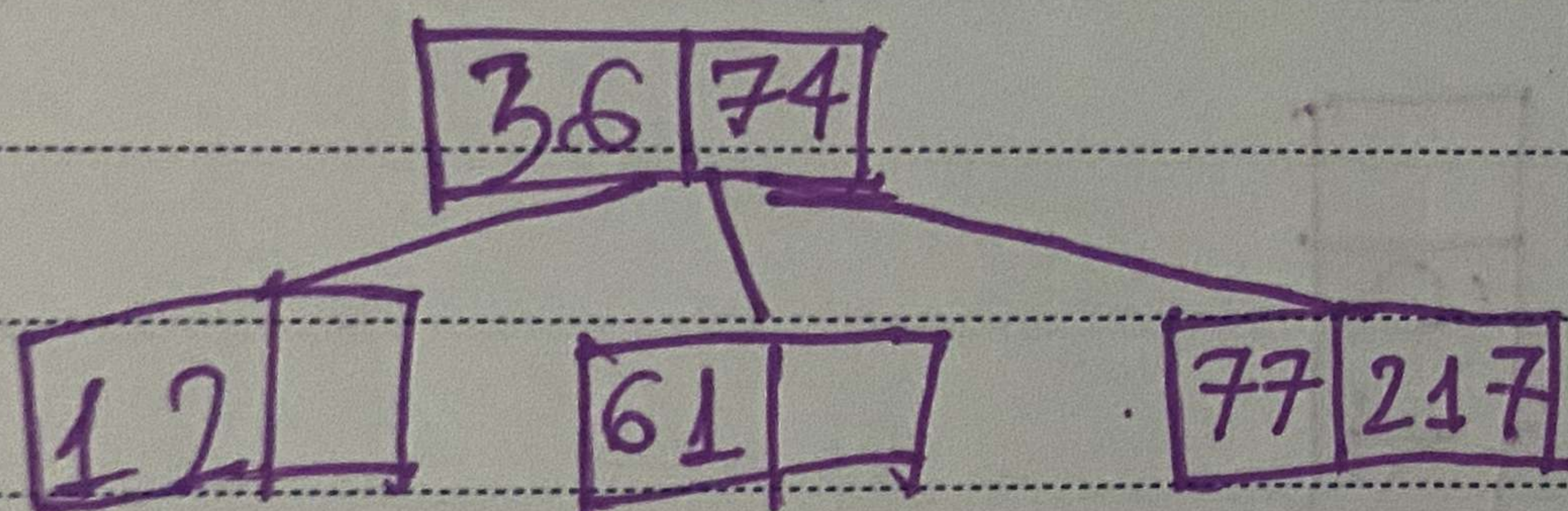
36 →



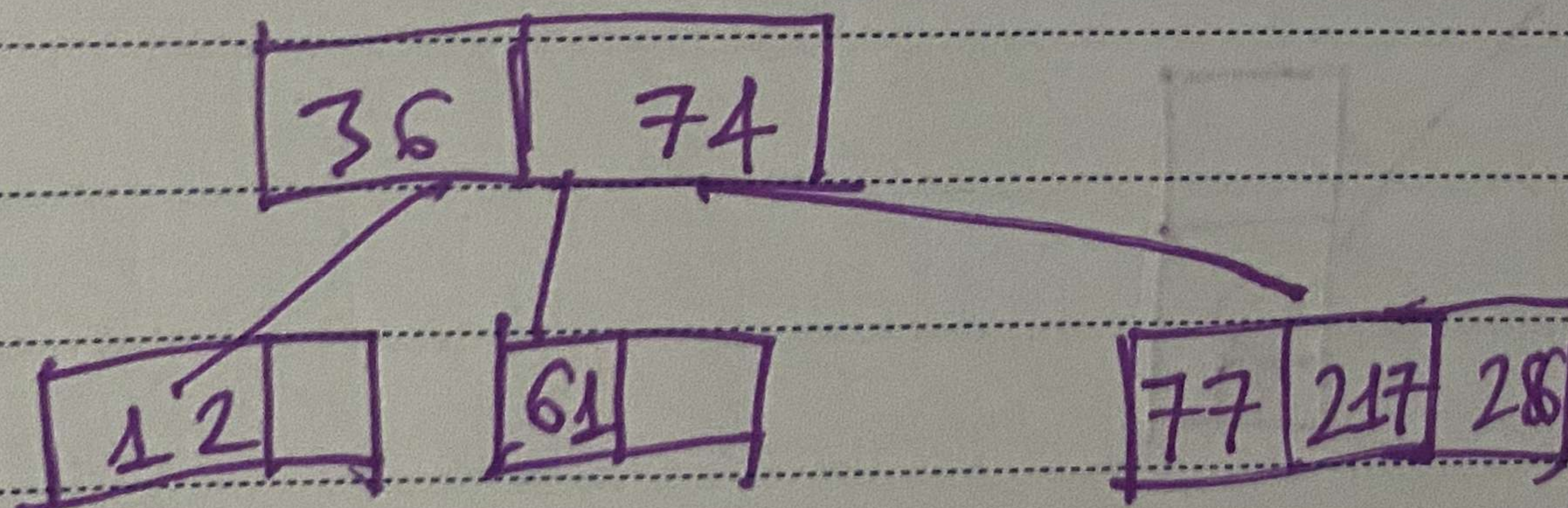
61 →



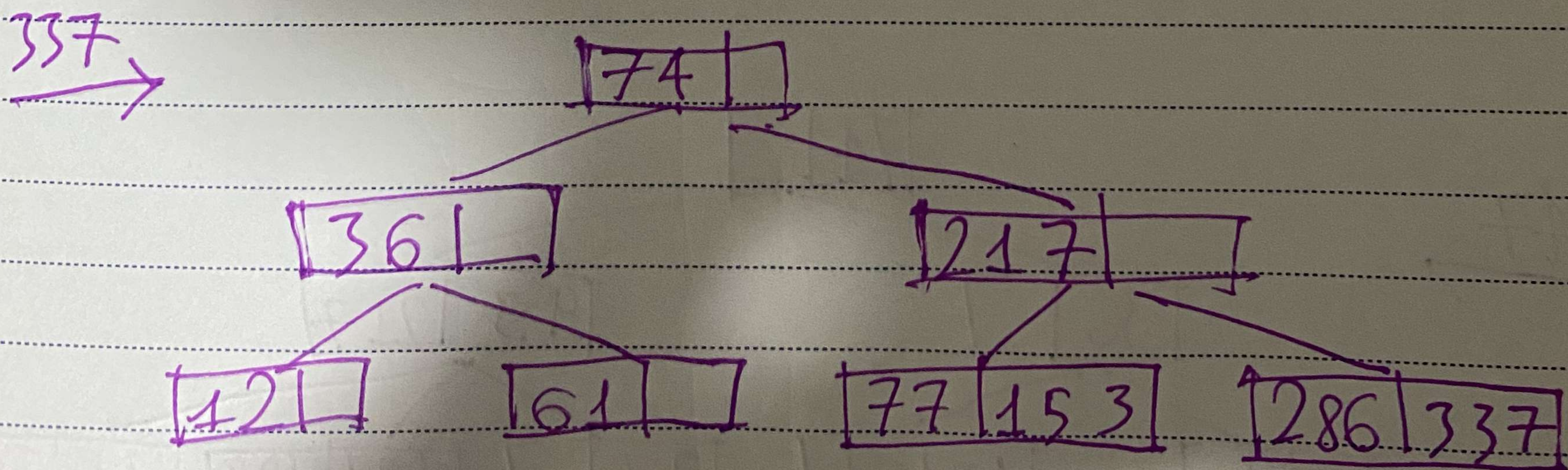
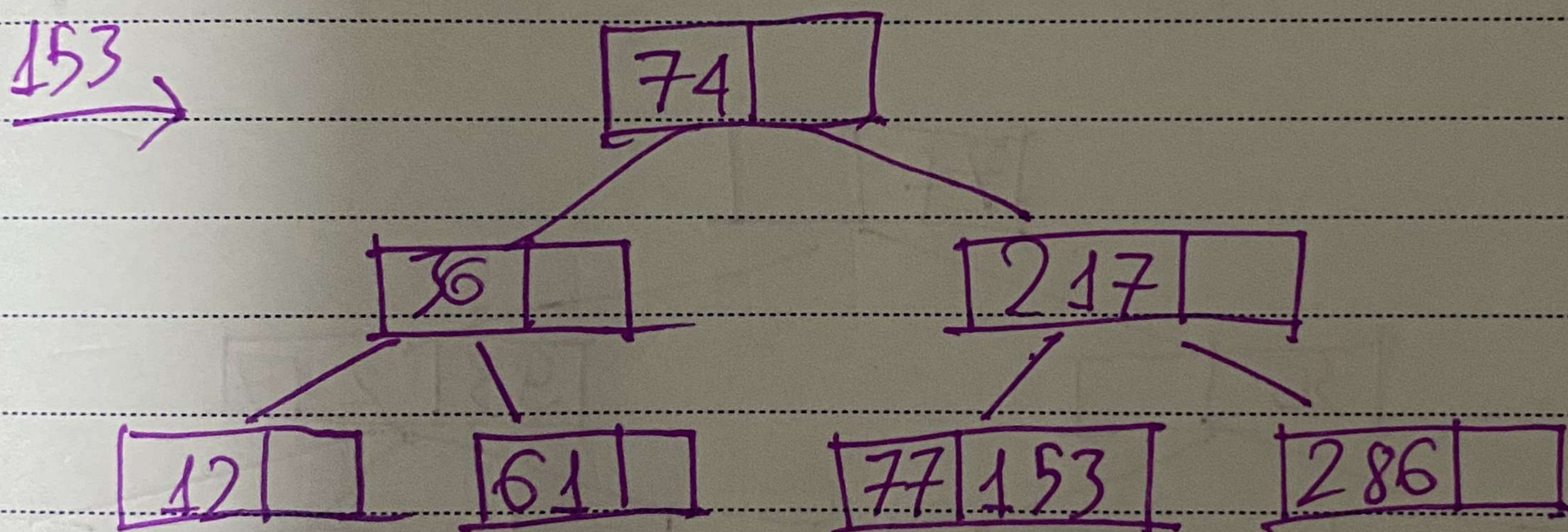
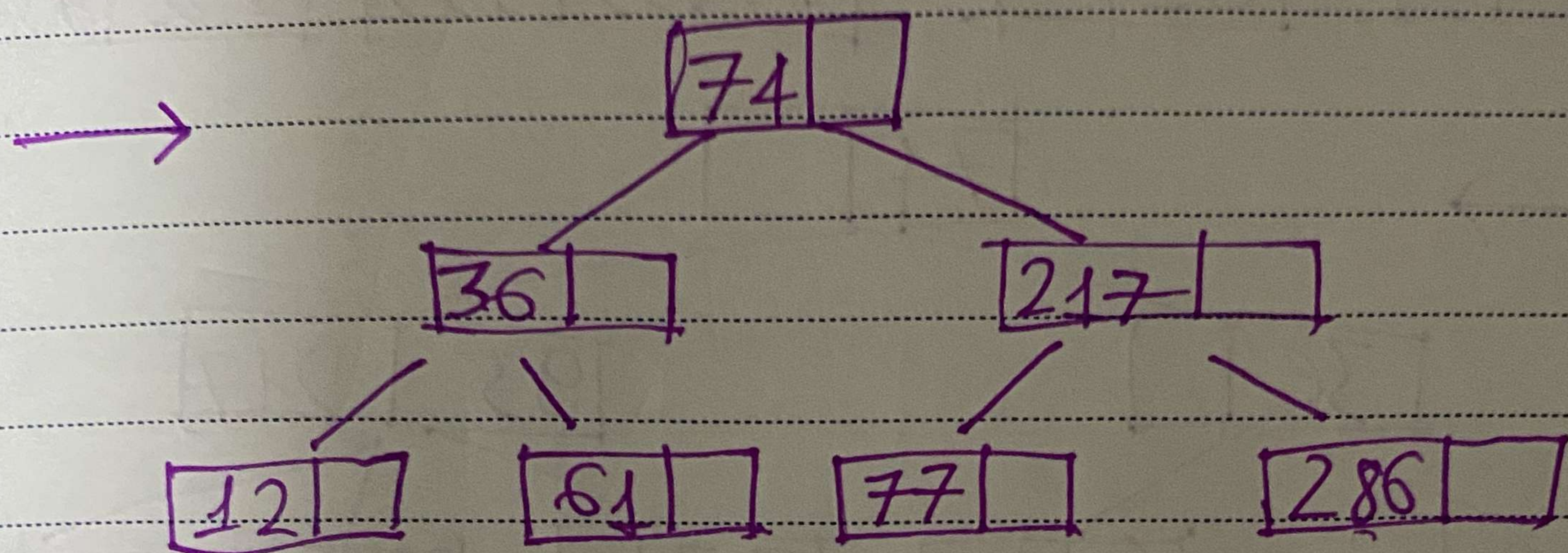
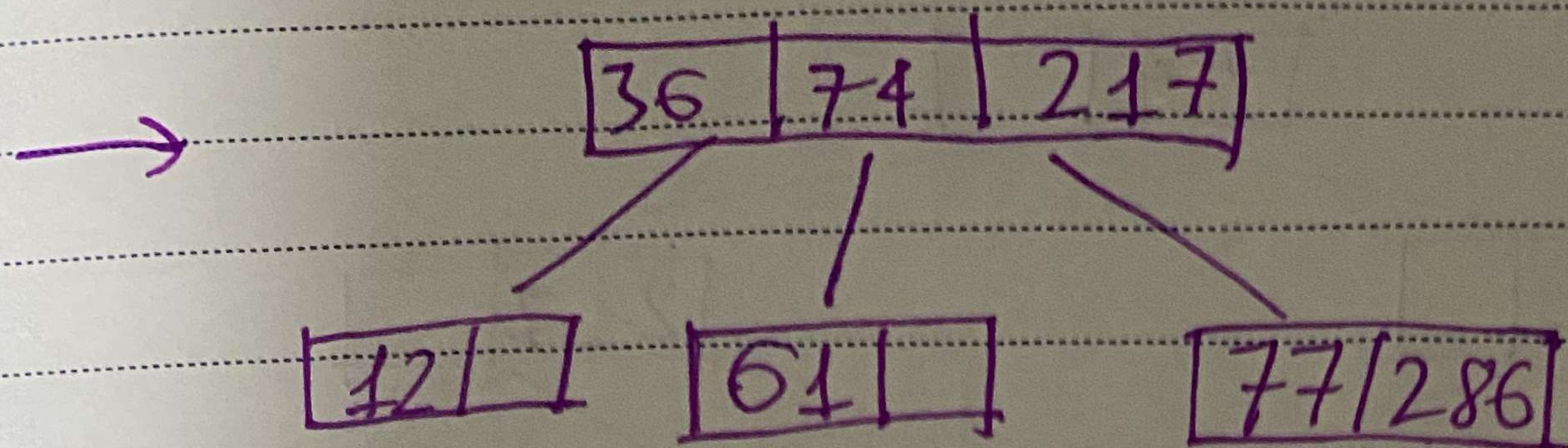
77 →



286 →



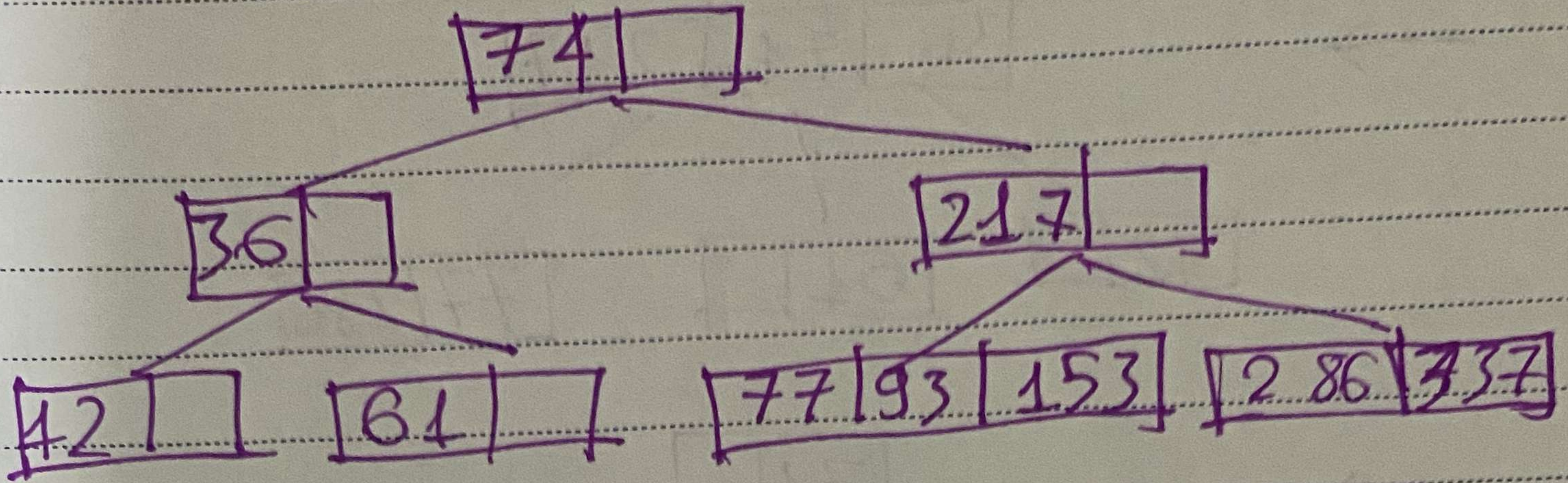




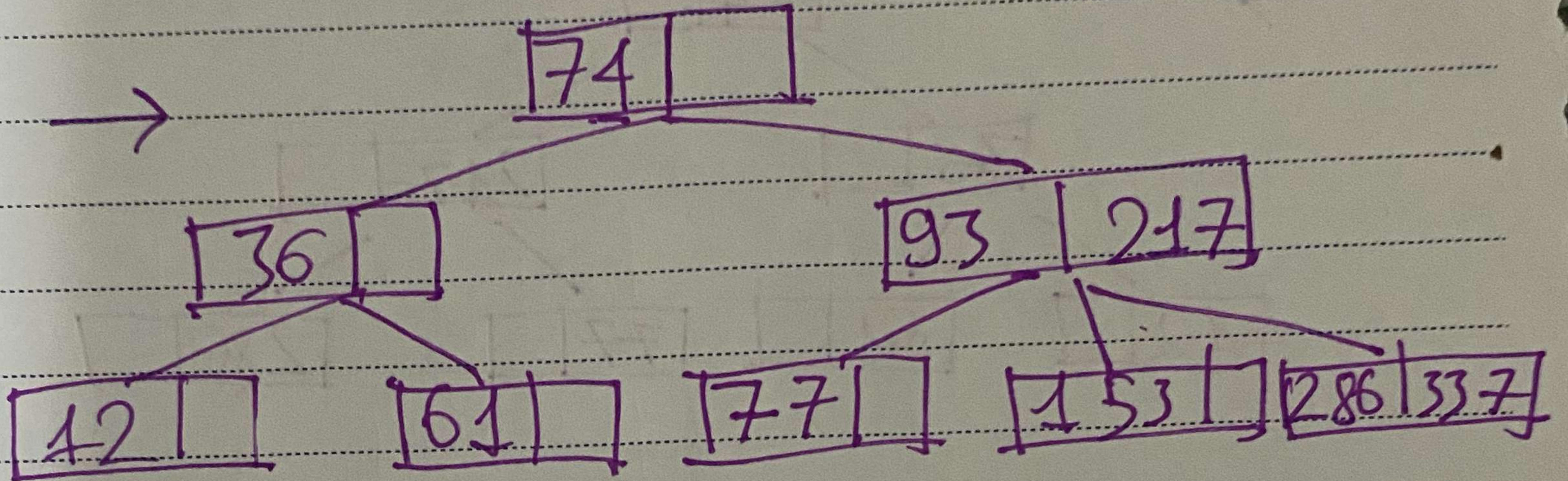


Date: ..... / ..... / .....

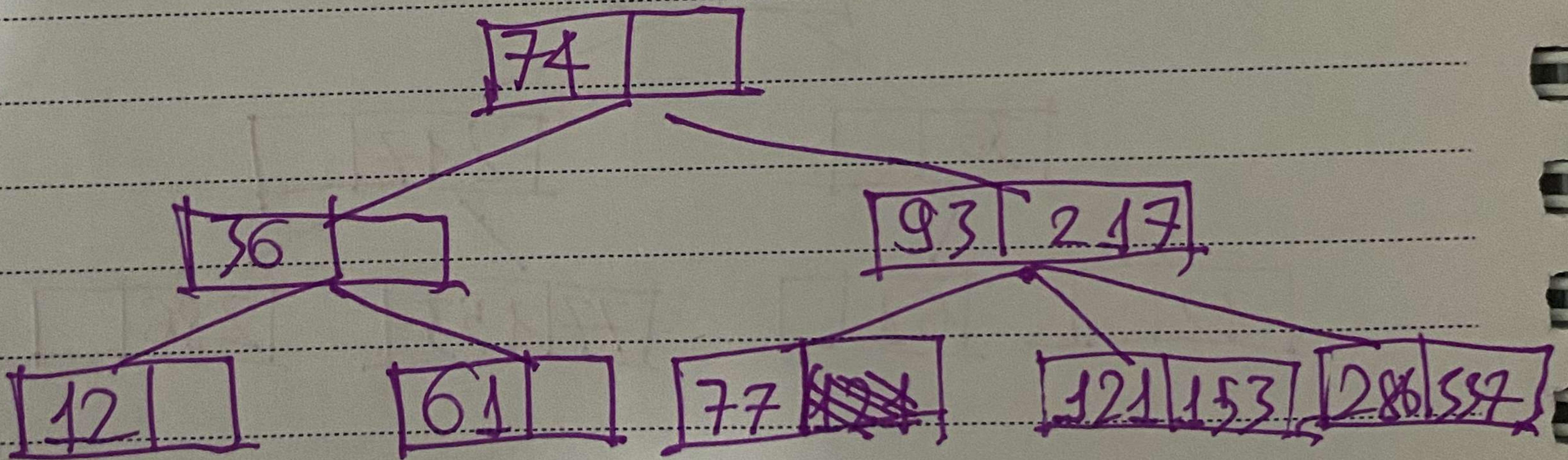
93 →



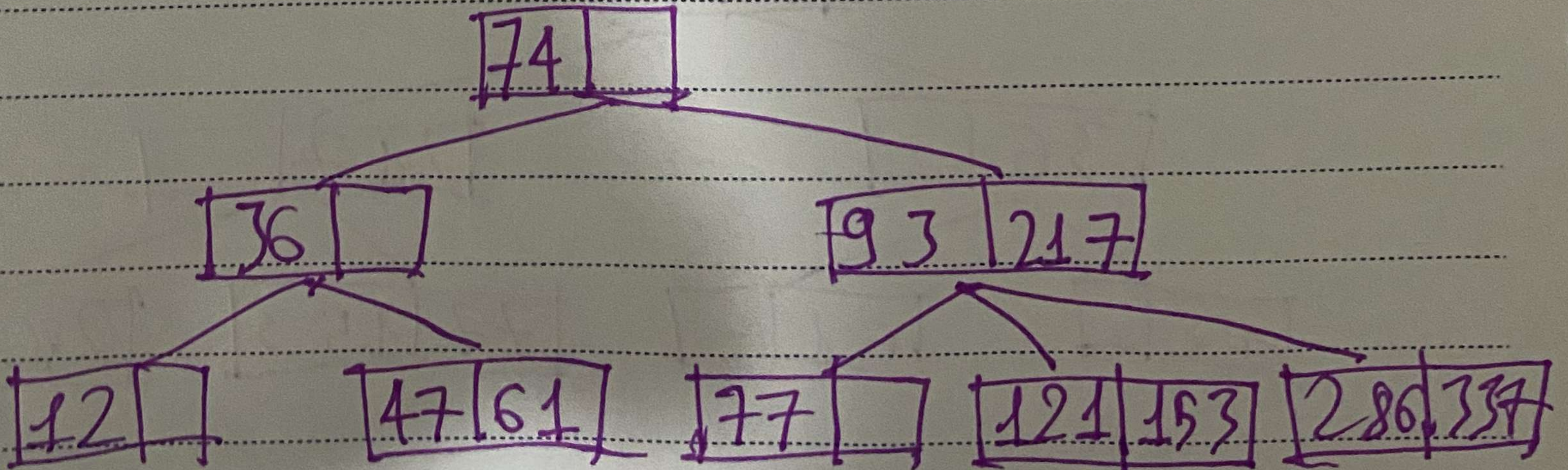
→



121 →

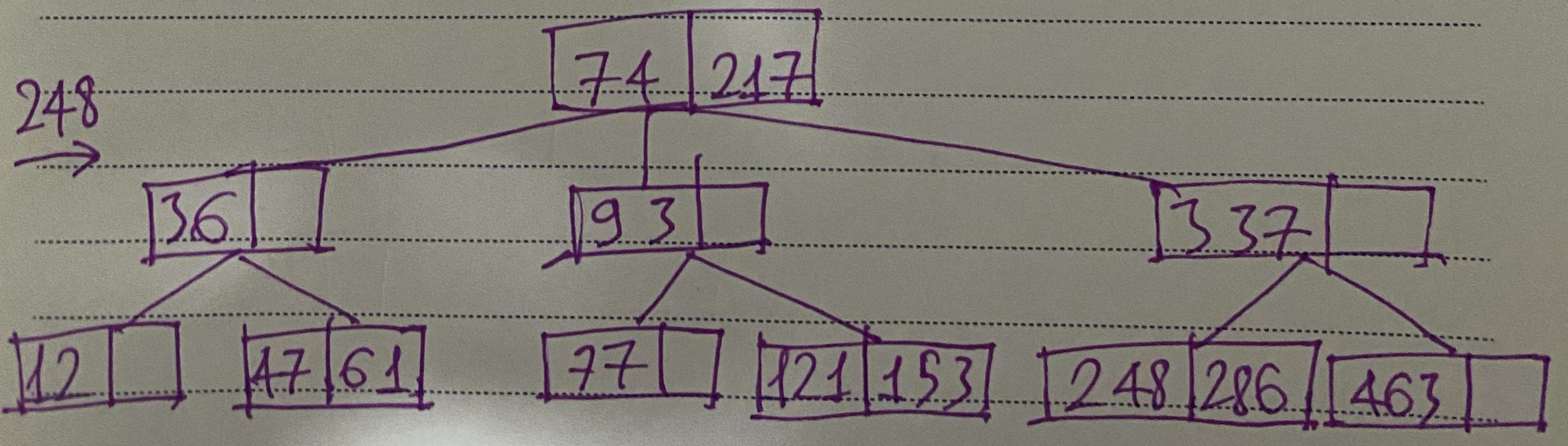
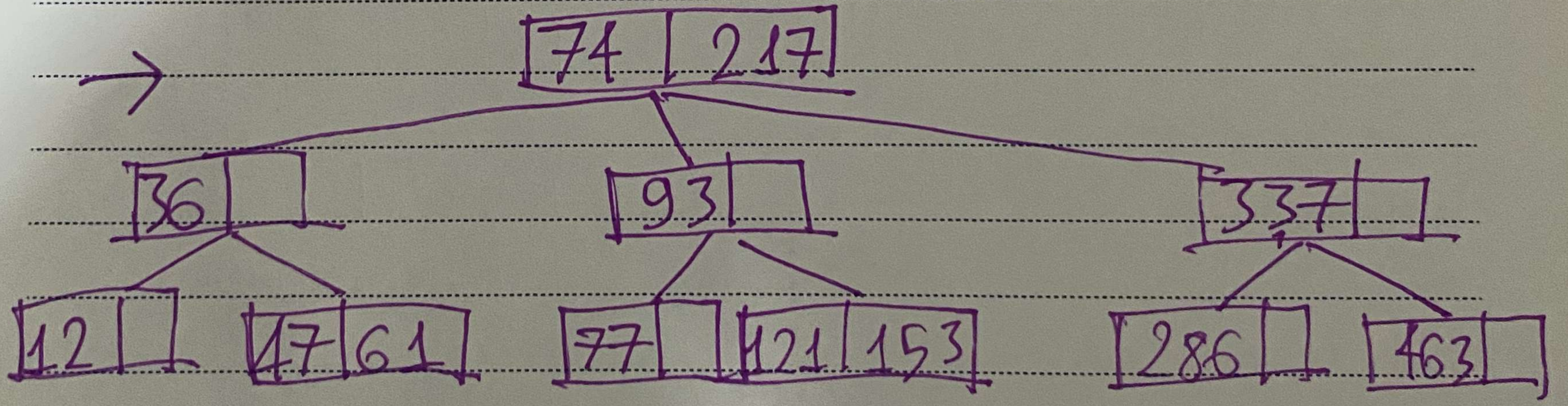
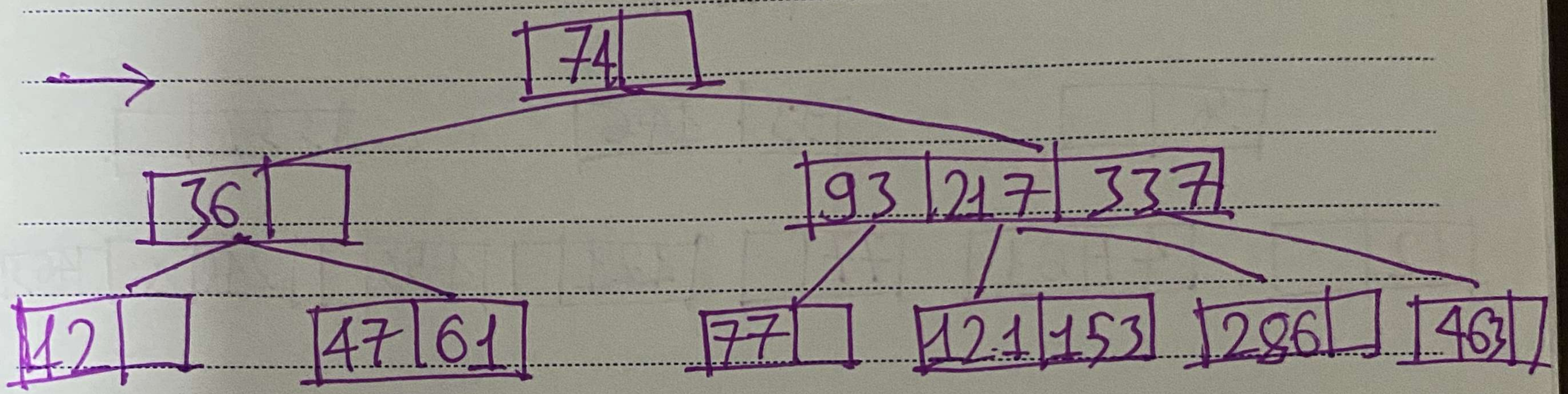
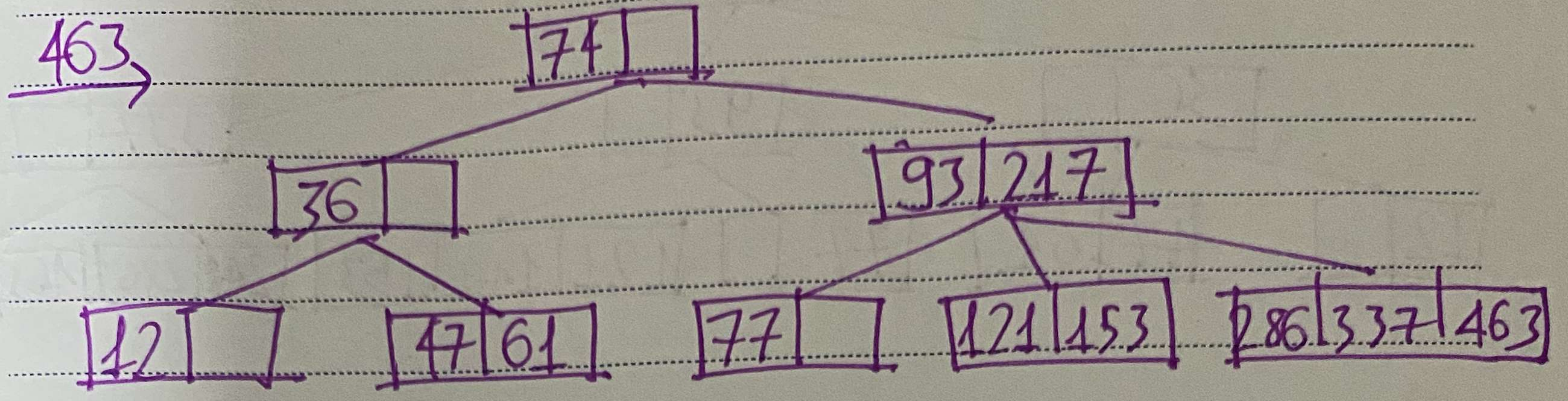


47 →





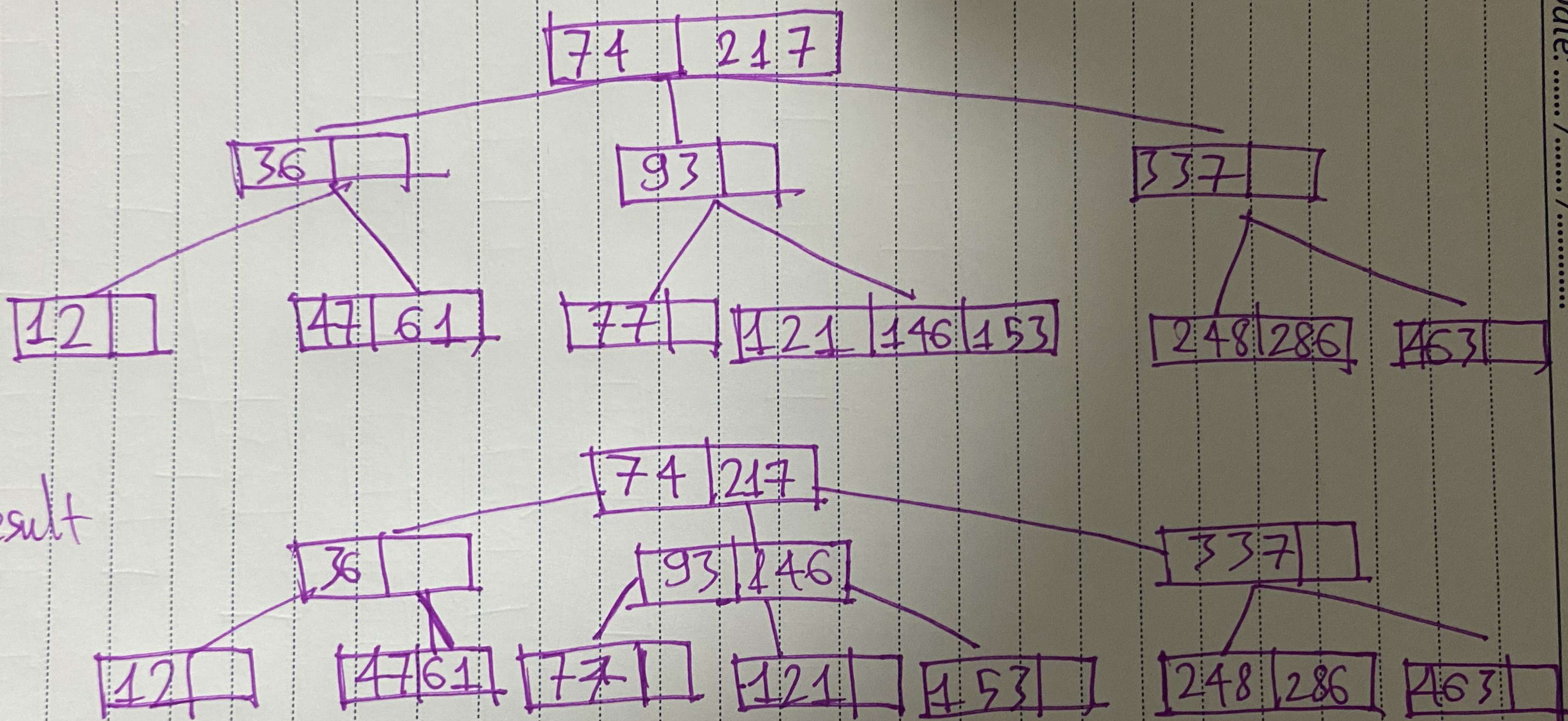
Date: ..... / ..... / .....





Date: ...../...../.....

fit@hcmus



146  
→

Final result  
→

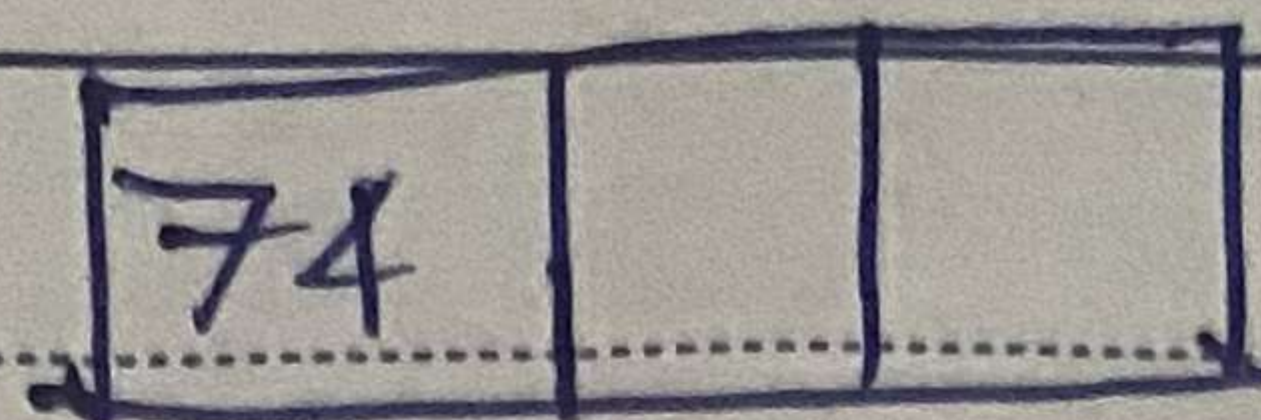


# Insert to 2-3-4 tree

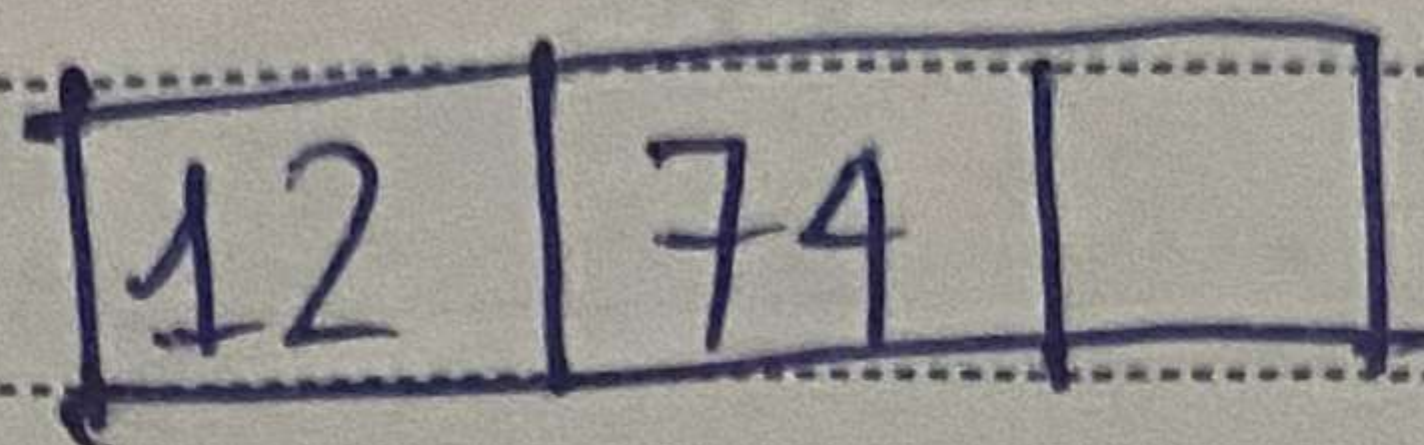
Date: ..... / ..... / .....

fit@hcm

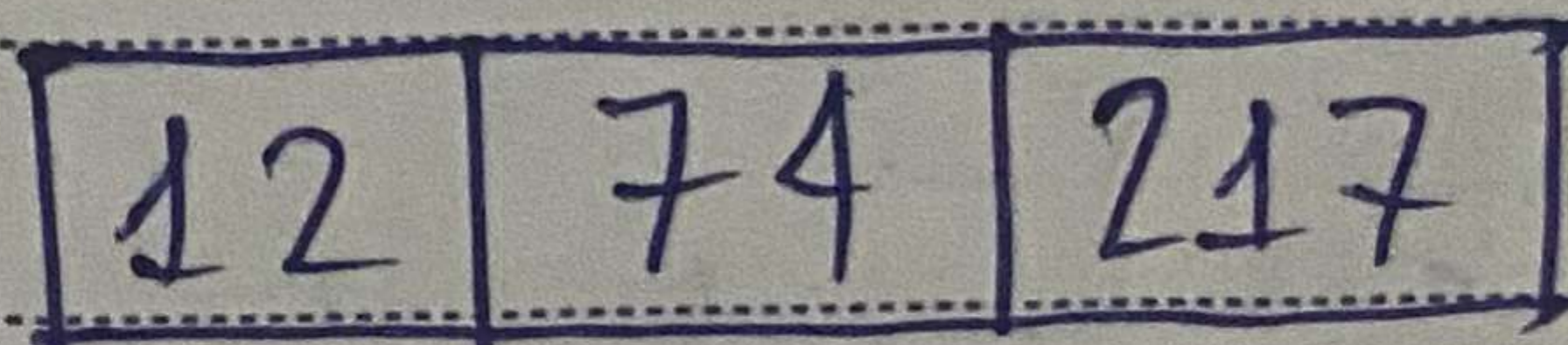
74 →



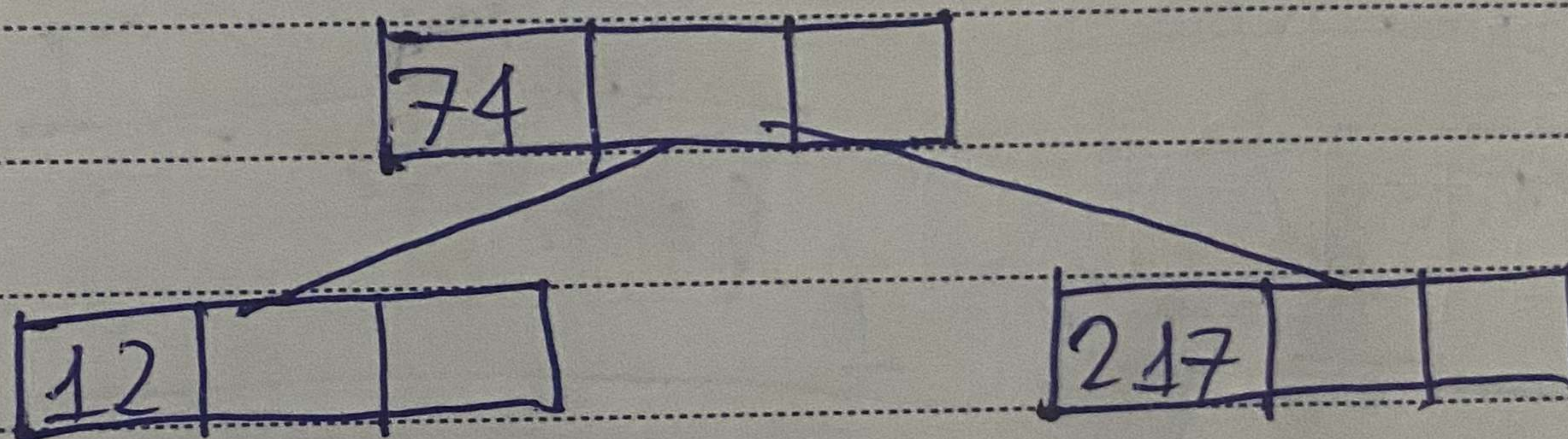
12 →



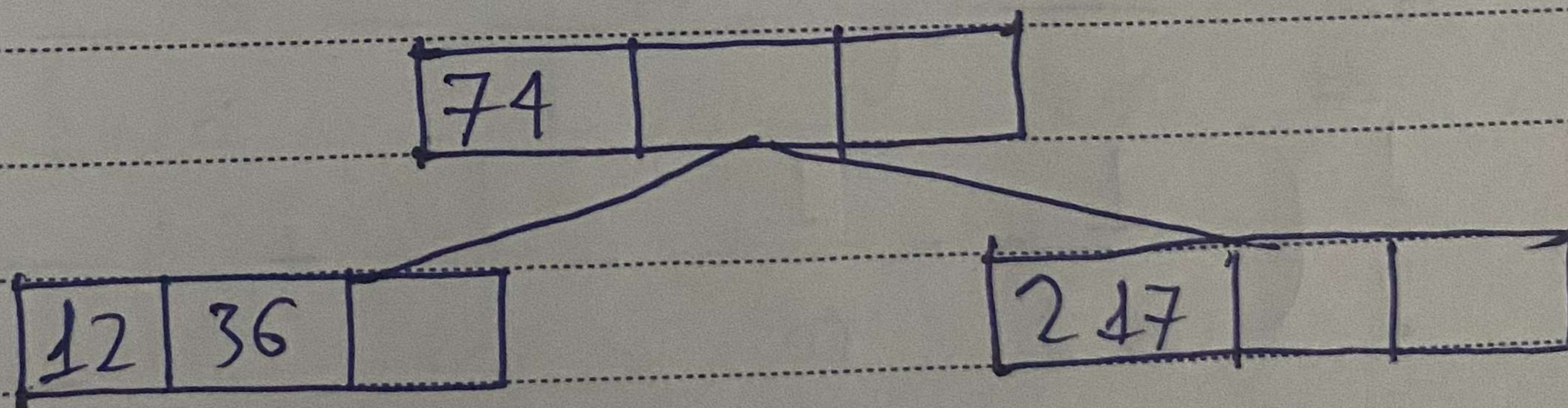
217 →



split →

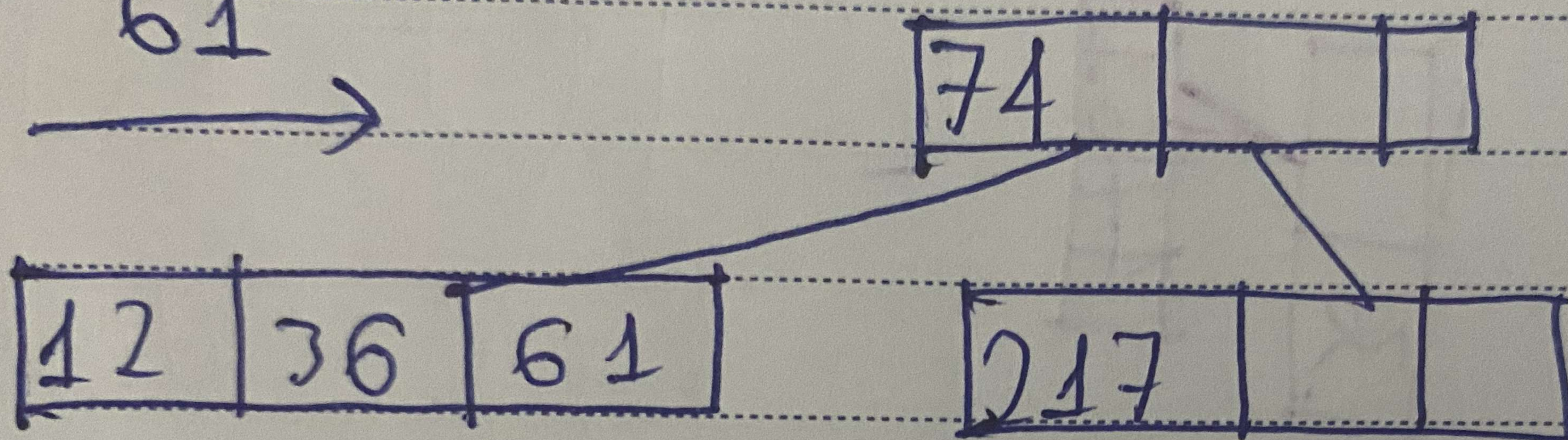


36 →

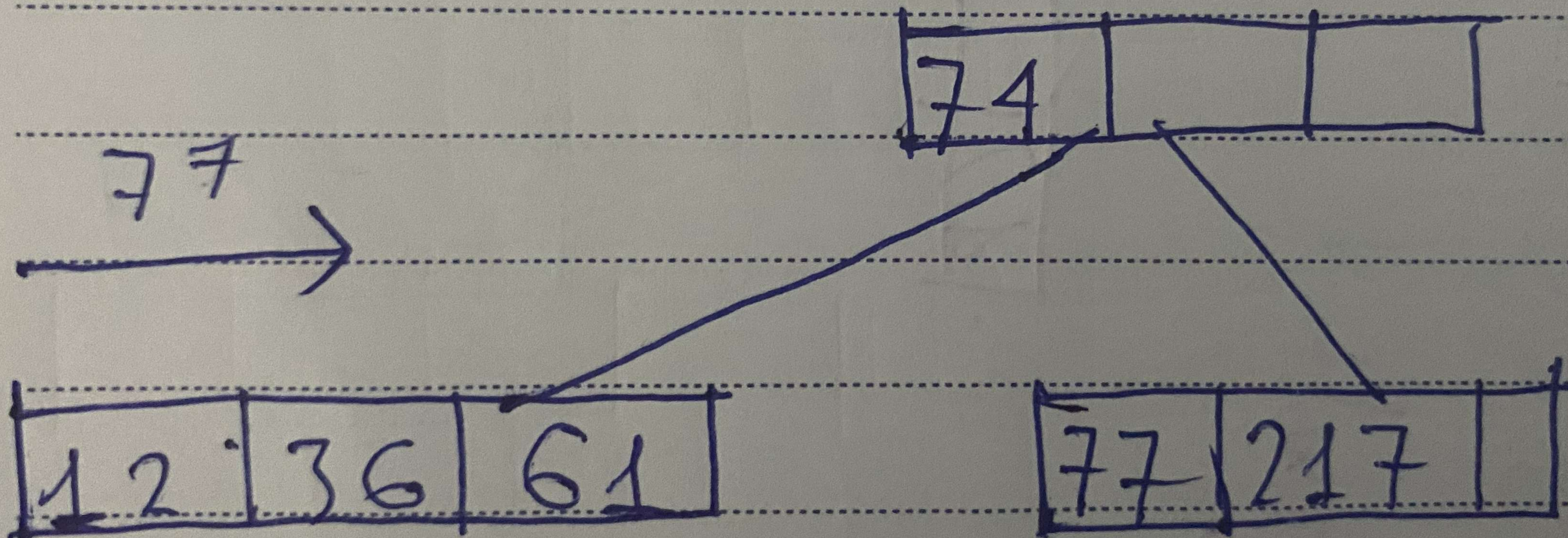




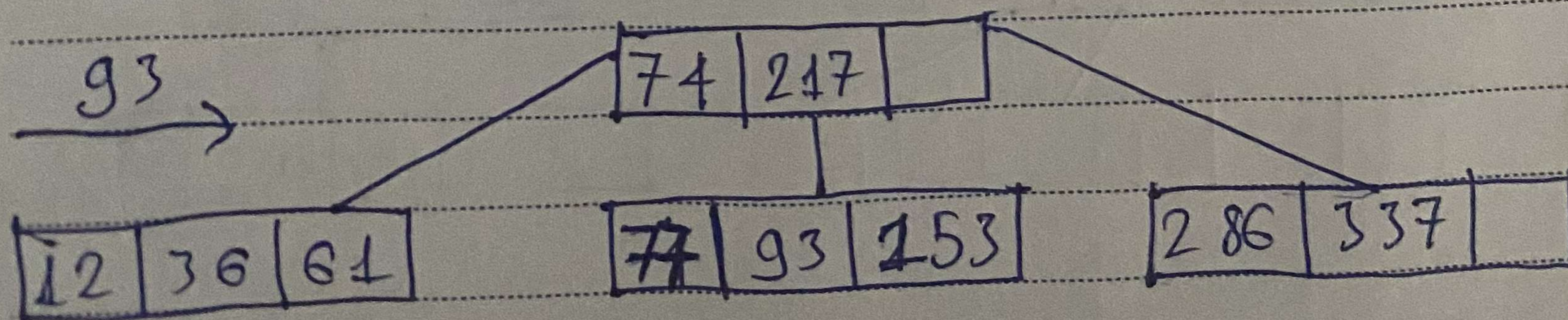
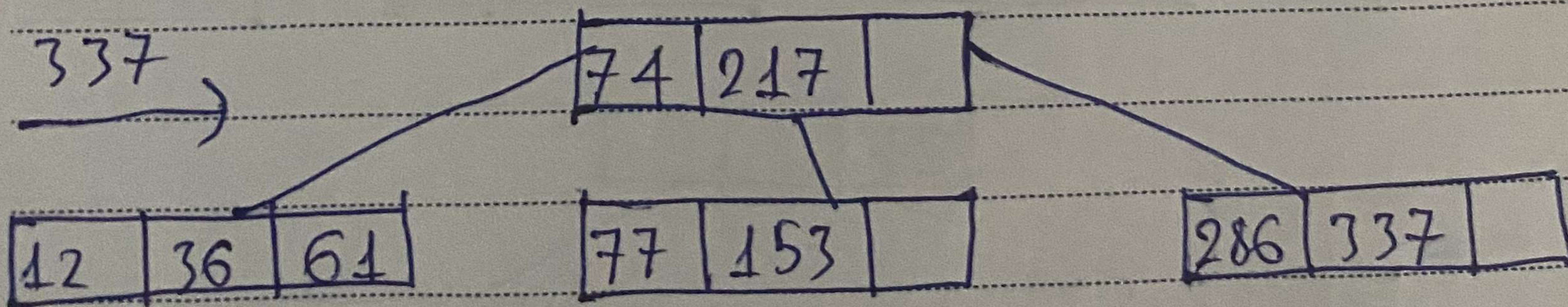
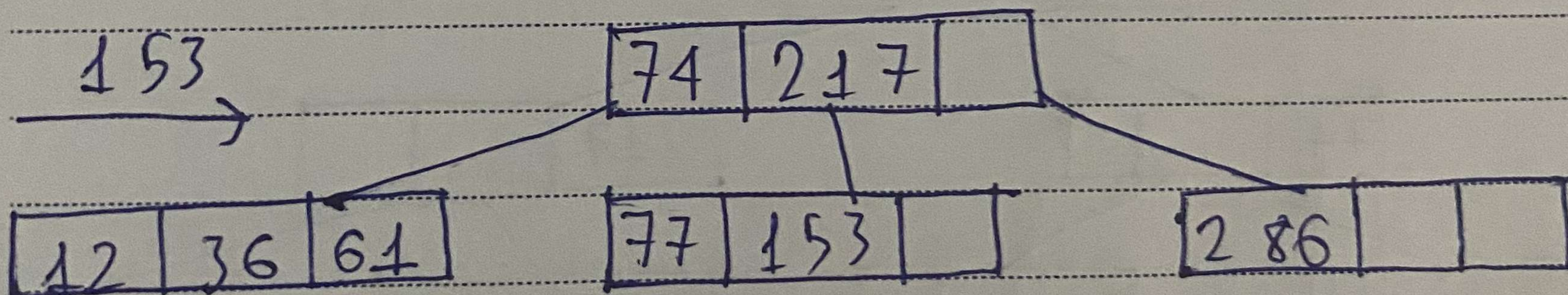
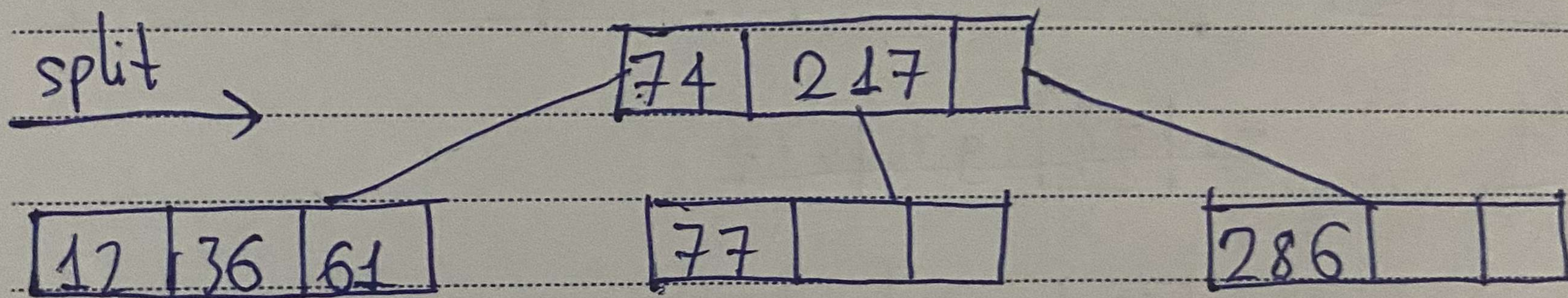
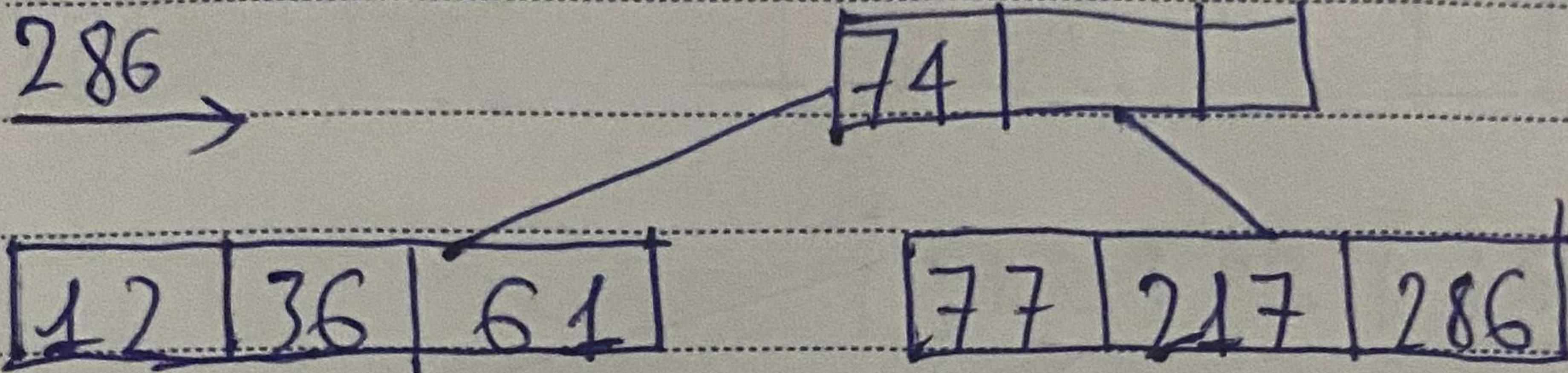
61  
→



77  
→

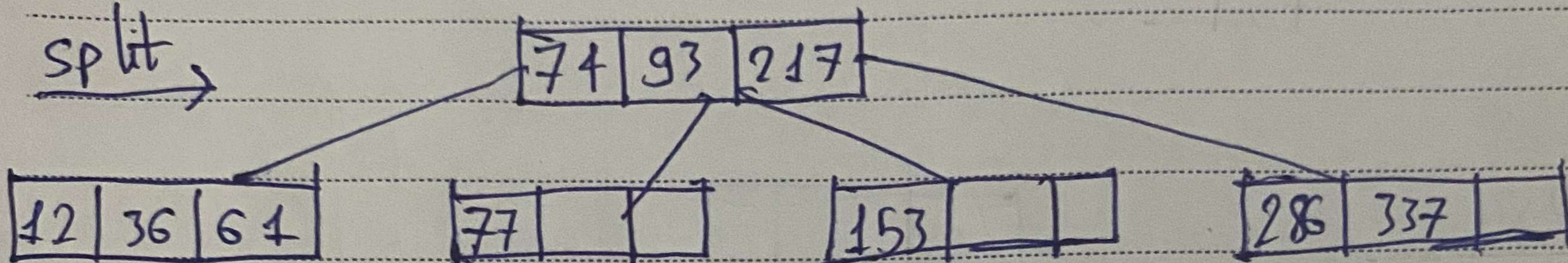




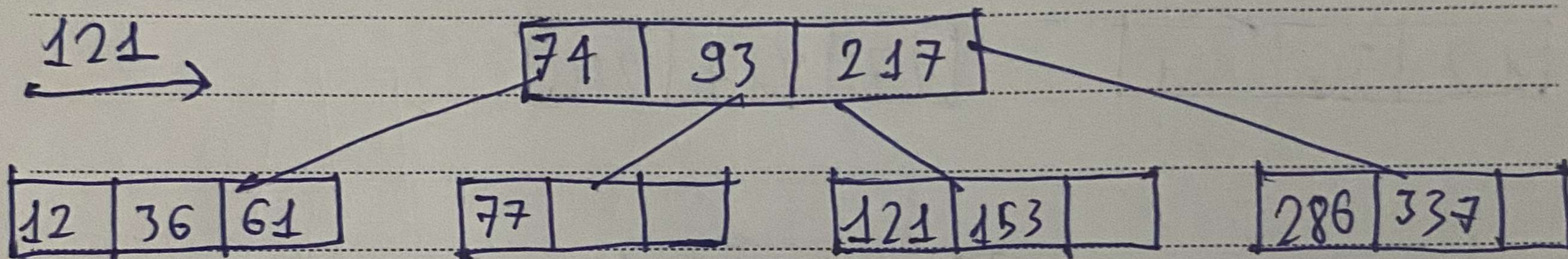




split →

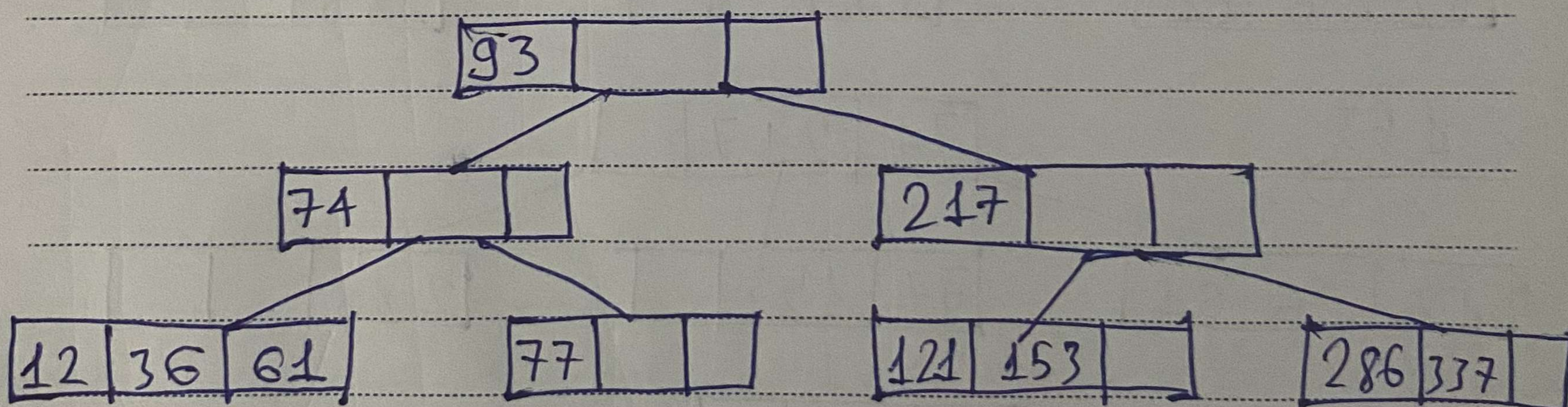


121 →

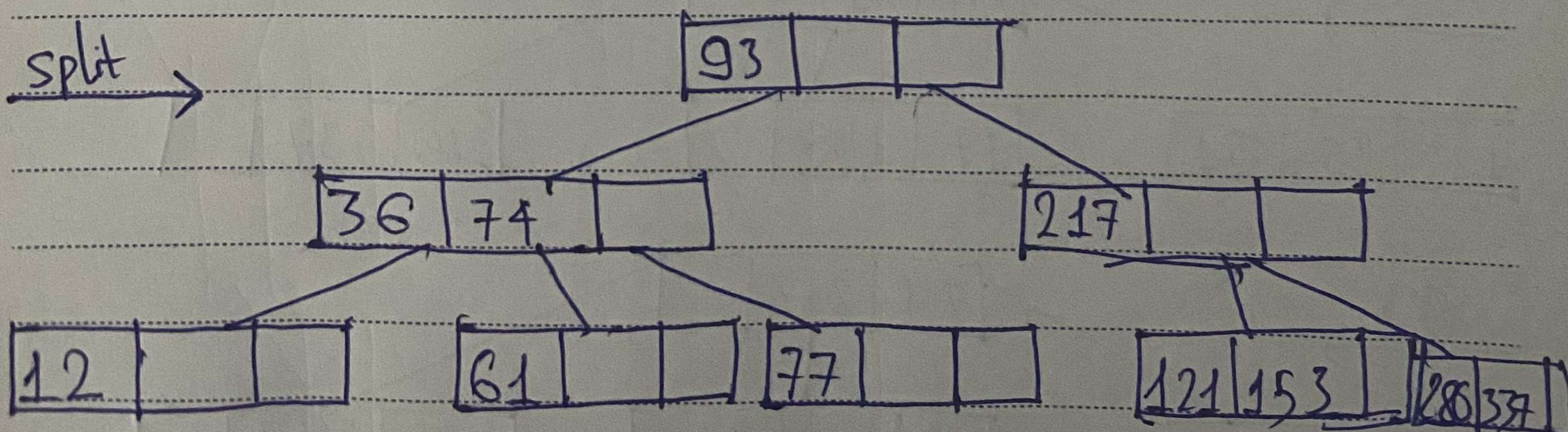


split →

36 | 74 | 93 | 217



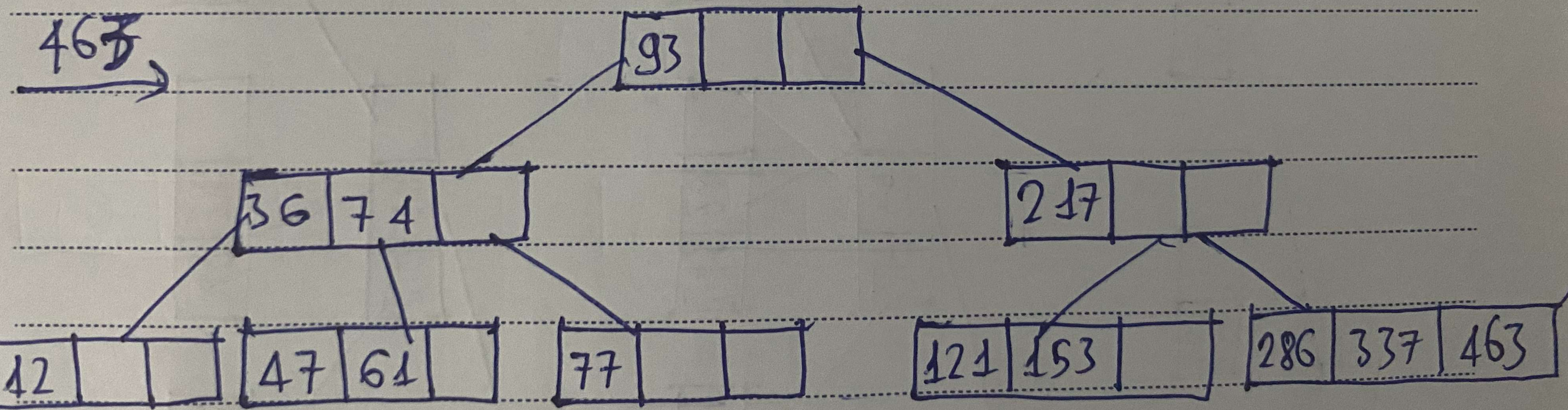
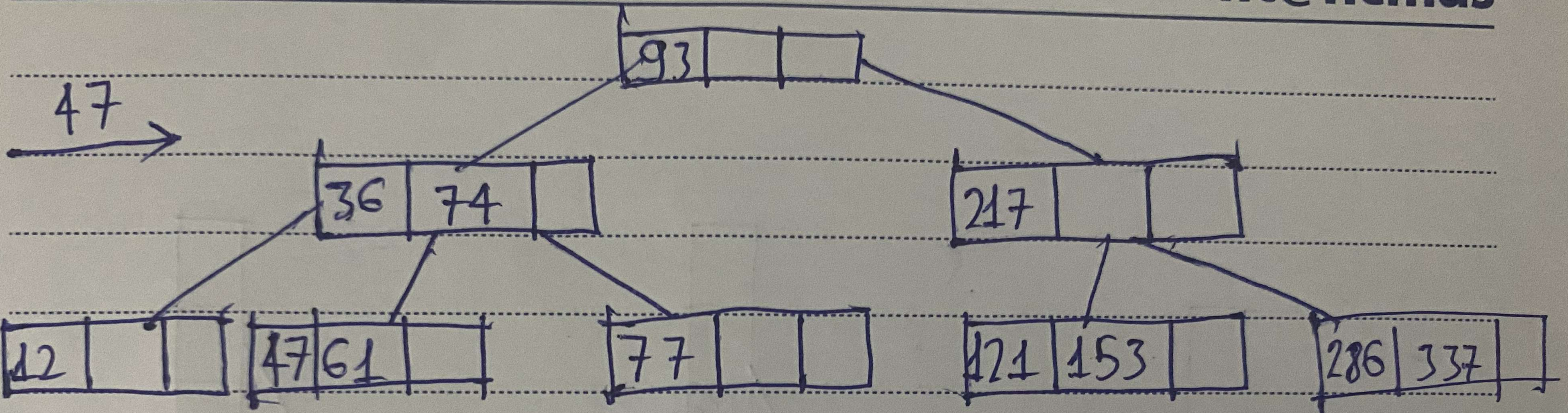
split →





Date: ..... / ..... / .....

fit@hcmus





split

248

146

